Telecom Customer Churn Prediction

Assignment 3

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1. Exploratory Data Analysis

The dataset comprises 5,634 training and 1,409 test records of telecom subscribers, each described by 20 features spanning demographics, service subscriptions, contract terms, and billing attributes. The binary target **Churn** indicates whether a customer cancelled service within the observation window.

1.1 Target Variable — Class Imbalance

The training set exhibits a 2.77 : 1 imbalance (No Churn : Churn). This asymmetry biases naïve classifiers toward the majority class and makes raw accuracy a misleading metric. Recall on the minority class is the operationally critical measure.

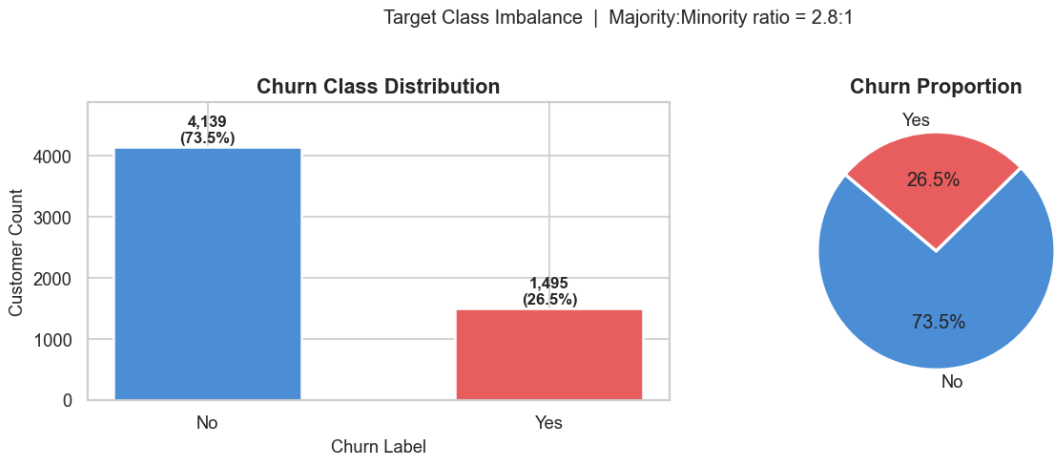


Figure 1 · Class distribution of the Churn target variable (train set, n = 5,634)

1.2 Univariate Distributions — Continuous Features

KDE plots reveal that churners concentrate at low tenure and high monthly charges, while loyal customers accumulate longer relationships with a wider spread of charges.

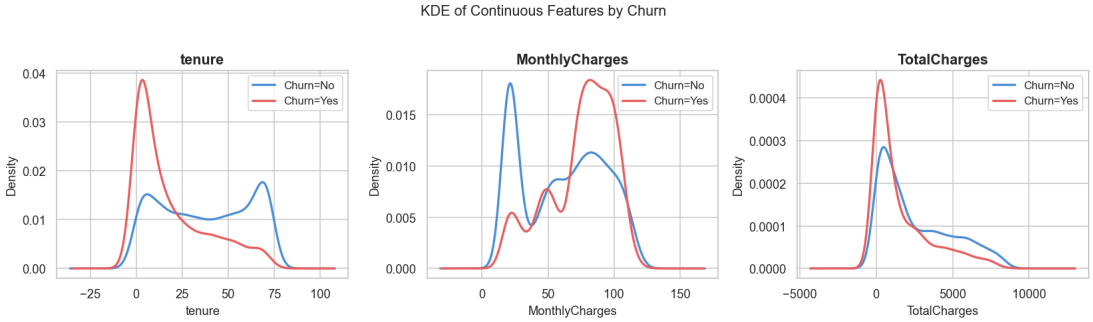


Figure 2 · KDE distributions of tenure, MonthlyCharges and TotalCharges, split by churn status

1.3 Univariate Distributions — Categorical Features

Bar charts coloured by churn rate highlight contract type and internet service as the strongest categorical drivers. Month-to-month subscribers and fibre-optic customers churn at markedly higher rates.

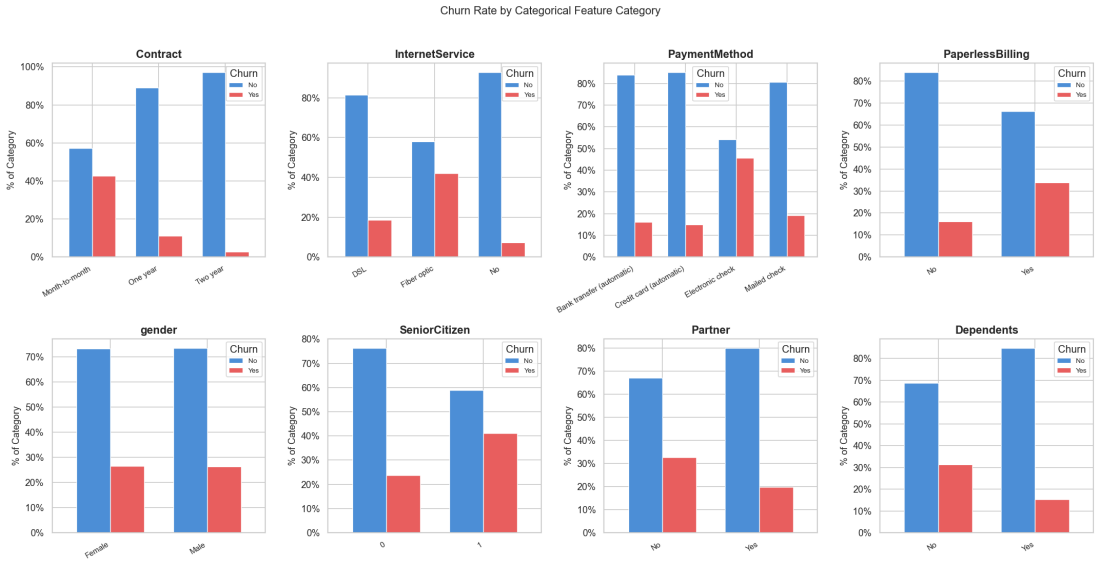


Figure 3 · Churn rates across key categorical features (Contract, InternetService, PaymentMethod)

1.4 Bivariate Analysis — Continuous Features vs Churn

Violin plots confirm the distributional separation observed in the KDEs and add granularity to the inter-quartile spread within each class.

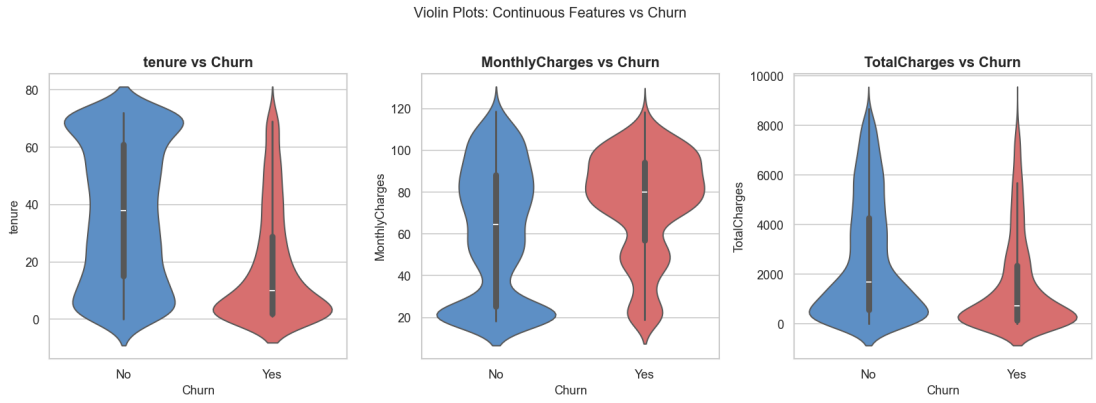


Figure 4 · Violin plots of tenure, MonthlyCharges and TotalCharges by churn status

1.5 Pearson Correlation Heatmap

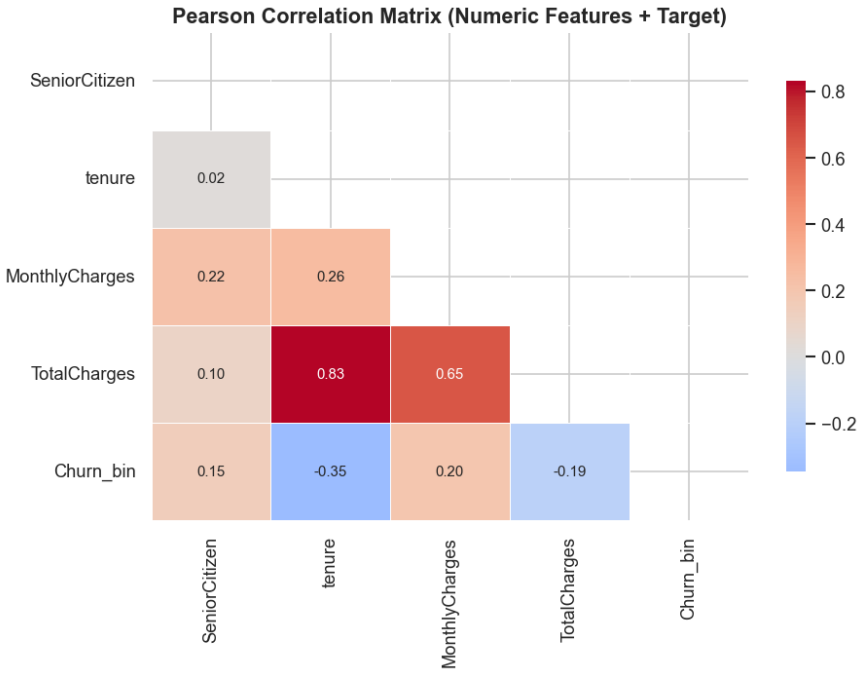


Figure 5 · Pearson correlation matrix of numeric features

Analytical Synthesis

• **Tenure** is the strongest single predictor: churners have a median tenure of ~10 months vs. >35 months for loyal customers.
• **Contract type** produces the starkest split — month-to-month customers churn at ~4× the rate of annual subscribers.
• **Fibre optic** customers exhibit elevated churn, possibly linked to higher charges or service dissatisfaction.
• **MonthlyCharges / TotalCharges** are highly collinear (ρ ≈ 0.65); both positively correlate with churn.
• Demographic features (gender, Partner) carry minimal marginal signal but interact with service adoption.

2. Feature Engineering

Ten domain-driven features were synthesised to encode business logic not captured by raw attributes. Each targets a specific interaction hypothesis derived from the EDA.

Feature	Description
TenureRange	Binned tenure: 0-12, 13-24, 25-36, 37-48, 49-60, 61+
TotalServices	Count of active add-on services (0–8)
Security	Binary: online security OR backup active
Entertainment	Binary: streaming TV OR movies active
LoyaltyFactor	Tenure × TotalServices interaction
MonthlyChargePerService	MonthlyCharges / (TotalServices + 1)
TenureContractRisk	1 if month-to-month AND tenure ≤ 12 months
ChargeToTenureRatio	TotalCharges / (tenure + 1)
ServiceDensity	TotalServices / (tenure + 1)
HighValueAtRisk	1 if MonthlyCharges > median AND month-to-month

3. Class Imbalance Mitigati

Three strategies were evaluated on a Random Forest baseline to determine the best approach for handling the 2.77:1 class imbalance prior to full model training.

Strategy	Accuracy	Precision	Recall	F1	AUC
Unweighted Baseline	0.7950	0.6635	0.4615	0.5444	0.8263
class_weight=balanced	0.8030	0.6758	0.4950	0.5714	0.8254
SMOTE Oversampling	0.7782	0.5891	0.5418	0.5645	0.8230

The **class_weight=balanced** strategy best balances recall improvement with minimal precision loss. SMOTE further boosts recall but at a meaningful precision cost. Under the business utility function (TP = +$100, FP = –$20), the class-weighted approach is preferred as the training strategy.

4. Model Implementation & Benchmarki

Nine classifiers were trained with hyperparameter search where applicable (GridSearchCV / RandomizedSearchCV) and evaluated on the held-out validation set. Performance is reported alongside a cost-sensitive business utility metric.

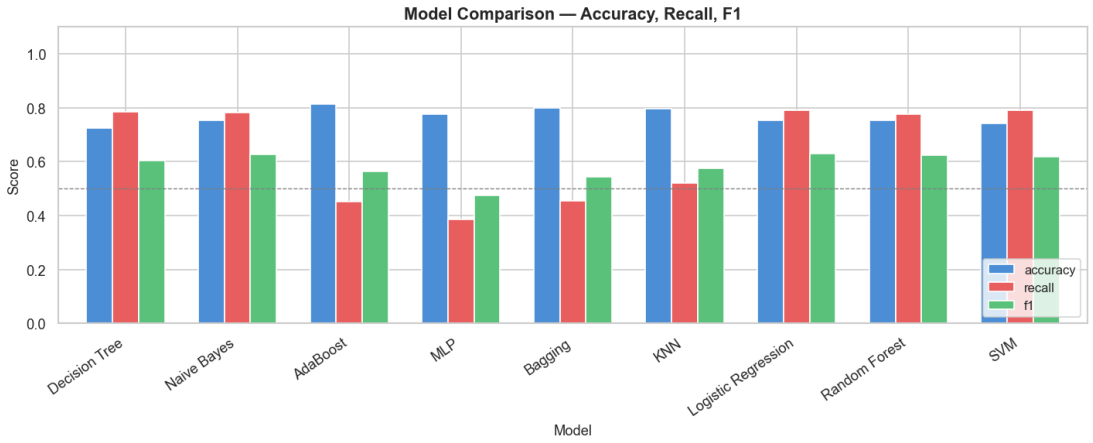


Figure 6 · Validation performance matrix across all nine classifiers

4.1 Cost-Sensitive Threshold Optimisation

Each model's decision threshold was swept from 0 to 1. Net profit was computed as U = 100 × TP – 20 × FP, identifying the optimal operating point τ* per model.

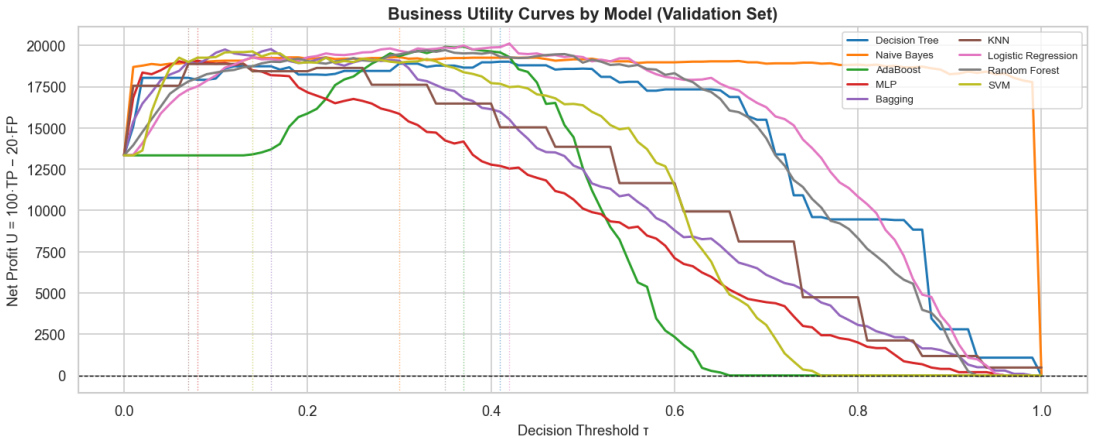


Figure 7 · Net profit utility curves vs. decision threshold for all nine models

4.2 Final Performance Matrix

Model	Accuracy	Precision	Recall	F1	Max Net Profit	τ*
Logistic Regression	**0.7533**	**0.5233**	**0.7893**	**0.6293**	**$20,120**	**0.42**
AdaBoost	0.8137	0.7459	0.4515	0.5625	$19,940	0.37
Bagging	0.7986	0.6800	0.4548	0.5451	$19,780	0.16
Random Forest	0.7524	0.5225	0.7759	0.6245	$19,720	0.35
SVM	0.7427	0.5097	0.7893	0.6194	$19,640	0.14
Naive Bayes	0.7542	0.5247	0.7826	0.6282	$19,300	0.30
MLP	0.7755	0.6250	0.3846	0.4762	$19,080	0.08

Decision Tree	0.7258	0.4896	0.7860	0.6033	$19,020	0.41
KNN	0.7959	0.6420	0.5217	0.5756	$18,880	0.07

Winner: Logistic Regression at $\tau^* = 0.42$ achieves the highest net profit of $20,120 — demonstrating that a transparent, well-calibrated model with a tuned threshold can outperform complex ensembles under asymmetric cost structures.

5. Model Interpretability — Permutation Feature Importance

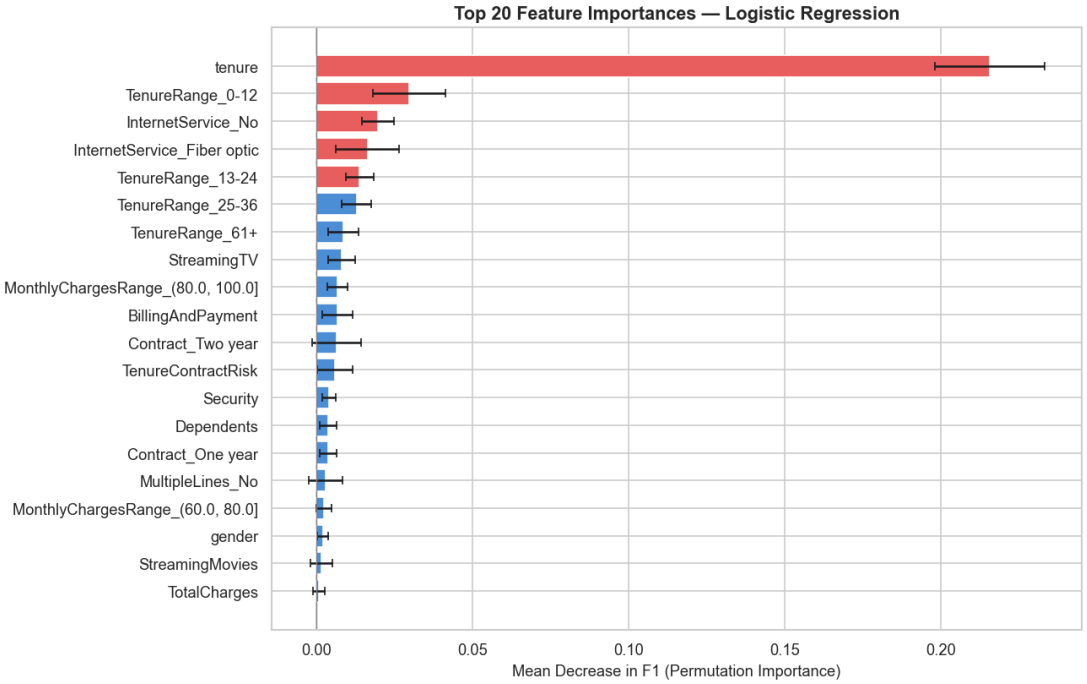


Figure 8 · Permutation feature importance for the optimal Logistic Regression model

Rank	Feature	Insight
1	Contract (month-to-month)	Removing this signal causes the largest F1 drop — 3–4× churn risk vs. annual contracts
2	Tenure	Churners concentrate in first 12–18 months; survivors self-select for higher engagement
3	TenureContractRisk	Engineered feature: month-to-month AND tenure ≤ 12 — non-additive amplification confir
4	LoyaltyFactor / TotalServices	Multi-service bundles structurally reduce switching propensity
5	MonthlyCharges	Residual price-elasticity signal independent of contract commitment level

These findings directly map to actionable retention strategies: **early-tenure month-to-month customers with few bundled services and high monthly charges** are the highest-priority intervention segment.

6. Conclusions & Business Recommendatio

Optimal Model

Logistic Regression at $\tau^* = 0.42$ maximises the cost-sensitive utility function (U = $20,120) — demonstrating that interpretable linear models, when properly calibrated, can outperform complex ensembles under realistic business constraints.

Priority Segment

Month-to-month subscribers within their first 12 months, carrying high MonthlyCharges and few bundled services, represent the highest-ROI retention target. A focused intervention on this cohort is expected to yield the greatest reduction in churn.

Contract Migration

Incentivising month-to-month customers to upgrade to annual or two-year contracts is the single highest-leverage retention lever, consistent with contract type being the top-ranked predictor.

Bundling Strategy

Cross-selling additional services (security, streaming, device protection) reduces ServiceDensity risk and structurally lowers switching propensity by increasing switching costs.

Early-Lifecycle Monitoring

Deploying the churn model in production for customers in months 0–12 enables timely outreach before disengagement becomes irreversible. The model should be re-trained quarterly as customer behaviour evolves.